

MINISTRY OF EARTH SCIENCES (MOES) ■ NATIONAL CENTRE FOR POLAR AND
OCEAN RESEARCH (NCPOR)

AI-Enabled Antarctic Sea-Ice, Iceberg Trajectory, and Navigation Decision Support System

Deep Research & Solution-Direction Report



Problem Statement ID: **26059** • Theme: Transportation & Logistics • Category: Software
Smart India Hackathon (SIH) • Technical research foundation for the project proposal

Prepared by: SIH Team • Research support document (no application code) • September 8, 2026

Scope, Method and Integrity Note

This report is a *research deliverable* prepared to support the development of a Smart India Hackathon (SIH) solution for **Problem Statement 26059**. No application code is produced here. Throughout the document, three kinds of statements are distinguished:

- **Facts** — established scientific and engineering knowledge, accompanied by citations;
- **Research findings** — results reported by specific, identified papers, cited exactly as reported by their authors;
- **Recommendations / assumptions** — analytical judgement about what is most appropriate for an SIH build, clearly flagged as such.

Nothing is fabricated. Where a source could not be verified as accessible, this is stated explicitly rather than glossed over (see the reference-paper analysis in Section 6). The Antarctic-specific vs. Arctic / general (transferable) distinction is preserved wherever evidence is cited, so that no Arctic finding is silently presented as Antarctic proof. Reported performance figures are taken from the cited literature and are *not* claimed for our own (unbuilt) system.

Honesty note (read first). The proposed system is a **decision-support** tool, not an autonomous navigation controller. It is a research proposal, not a validated or safety-certified operational system. Claims about accuracy, safety, or novelty are limited to what the cited literature supports and are clearly labelled as proposed methodology, assumptions, or future work.

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1 Executive Summary

The problem. NCPOR — and, by extension, any Antarctic research operator — runs multi-week supply and science voyages between Cape Town and its stations **Maitri** (Schirmacher Oasis, Dronning Maud Land) and **Bharati** (Larsemann Hills, Prydz Bay), and across the Southern Ocean, **only in the austral summer (roughly November–March)**. These transits pass through the world’s most dynamic and least-observed sea-ice zone, are blocked or constrained by drifting sea ice, must avoid icebergs and bergy bits that are present all year round, and operate under sparse satellite / in-situ data and extreme weather. The SIH problem asks for a **decision-support platform**, not an autonomous controller, that fuses (1) a *forecast* of Antarctic sea-ice concentration, (2) *prediction of iceberg trajectories*, and (3) *safe and fuel-efficient route generation* into one system that a voyage-planner / navigator can act on.

The one-line problem. Route a research vessel safely and efficiently through a moving, uncertain field of Antarctic sea ice and icebergs — and support the human who must actually make that decision.

Key research findings that shape the solution:

- **Sea-ice concentration (SIC) forecasting is an established, rapidly maturing field.** Data-driven models (ConvLSTM, IceNet, hybrids) now demonstrably compete with or beat dynamical coupled-climate models at sub-seasonal-to-seasonal horizons in the SIPN / SIPN-South community comparisons (Massonnet et al. 2023; Dong et al. 2024). But *most Antarctic SIC deep-learning work targets basin-scale, monthly-to-seasonal forecasting for climate science — not the daily, ~7–15 day, route-region forecast a navigator needs*. That is a genuine gap.
- **Iceberg trajectory prediction is a separate, harder science** dominated by physical force-balance drift models (wind + current + Coriolis drag), plus recent hybrid physics+ML approaches. It is essentially never fused into the same system as SIC forecasting or route optimization. *This is the clearest integration gap*.
- **Ice-aware route optimization is mature in the Arctic / Northern Sea Route literature** (POLARIS-based risk, A*, Dijkstra, ant-colony, genetic, D* Lite, wave-front), and a UK (BAS) *PolarRoute* prototype exists for Antarctic. Almost all of it is published as offline method papers or Arctic tools; it is rarely Antarctic-specific and rarely ships the weather / ice *forecast* inside the route cost.
- **The three components (SIC forecast, iceberg forecast, route planning) live in different papers and different disciplines.** A combined “forecast + risk + optimize” Antarctic platform — evaluated on a defined route with a defined vessel and honest uncertainty — is a real, defensible integration / operational gap, not a manufactured one.
- **Data availability is workable for a prototype.** SIC products, ERA5 atmosphere / waves, ORAS5 / ECCO currents, IBCSO bathymetry, and Sentinel-1 SAR are all free / open and cover Antarctica. The scarce resources are (a) iceberg ground-truth tracks, (b) high-resolution validated current / wind *forecasts* in ice, and (c) vessel-specific ice-transit performance data.

Recommended architecture (summary). A hybrid, not pure-deep-learning, system:

1. **SIC forecasting:** start with a ConvLSTM / UNet-type spatiotemporal net trained on open SIC reanalysis / CDR fields (extend later with ERA5 / ORAS5 channels) to produce daily probabilistic SIC maps out to ~7–15 days over the relevant sectors, with uncertainty. This is the *only* component where deep learning is clearly justified and feasible.
2. **Iceberg drift:** a physics-based force-balance drift model (deterministic core) + ensemble / particle spread (uncertainty) + optional learned residual correction — following Wagner / Bigg-type equations and the IDRIFTNET philosophy — rather than a pure black-box regressor, because labelled iceberg tracks are scarce.
3. **Routing:** a graph (grid)-based, time-varying, multi-objective least-cost route search (Dijkstra / A* or wave-front, or D* Lite for dynamic re-planning) where each cell cost = weighted

(vessel transit / fuel penalty in ice) + (POLARIS-like ice risk) + (iceberg encounter risk) + (weather / time penalties), respecting a hard “no-go” constraint. A Pareto / weighting approach surfaces the safe route vs. the fuel-optimal route for the human to choose.

4. **Explainability & decision support:** uncertainty bands, per-segment risk explanations, alerts, alternate routes, and “what would change if the ice forecast shifts” sensitivity — this is what makes it a *decision support* system rather than a “black box that draws a line.”

Honest innovation claim. The system is *not* novel because it uses “AI,” and SIC forecasting or ice routing should not be claimed as new. The defensible contribution is an *integrated, Antarctic-route-specific decision-support pipeline that couples a short-horizon probabilistic sea-ice forecast with iceberg-drift prediction and vessel / ice-aware multi-objective routing under uncertainty, evaluated on real NCPOR-style routes and vessels, with explainable risk output*. The demonstrable differentiators are **integration + Antarctic / operational specificity + uncertainty + explainability**, each grounded in a body of research that treats these components in isolation (Sections 9–14).

2 Problem Understanding

2.1 The actual problem being solved

A vessel master / voyage-planner must decide, days to weeks ahead, **where to route a research vessel through a heterogeneous, time-evolving field of drifting sea ice and icebergs**, balancing three conflicting goals:

1. **Safety** — avoid besetment (becoming trapped) in heavy ice, avoid collision / damage from ice floes and icebergs, avoid dangerous weather.
2. **Progress / efficiency** — minimise time and **fuel** to reach a station or survey line.
3. **Mission** — reach the science area, maintain schedule windows dictated by the short season and by crew / station logistics.

A decision-support system (DSS) turns raw environmental fields into **forecasts, risk maps, candidate routes, and trade-off explanations** that a human decision-maker uses. It does *not* replace the captain.

2.2 Intended users (operational environment)

- **NCPOR voyage / marine-logistics planners** (Cape Town → Maitri → Bharati ↔ and back; ~10–12 days open ocean + 5–7 days ice-affected between stations in the season).
- **Bridge officers / masters** of ice-capable chartered vessels and of India’s planned indigenous polar research vessel.
- **Ice-pilot / science cruise planners** choosing survey lines in the Southern Ocean.

Secondary users: scientists needing SIC / iceberg situational awareness.

2.3 Expected inputs → outputs

- **Inputs:** SIC fields (observed + forecast), iceberg detection / tracks, ocean currents, wind, waves, temperature, bathymetry, vessel characteristics (ice class / speed–power), planned origin / destination / departure.
- **Outputs:** SIC forecast maps with confidence; iceberg trajectories with uncertainty cones; navigability / risk maps; a recommended route and alternates; ETA; fuel estimate; risk and rationale per segment; alerts; “what-if.”

2.4 Constraints (operational reality — see Section 12)

Only-Antarctic-summer navigation window; ice-class limits; limited connectivity and bandwidth; sparse / repeat-pass satellite coverage; inability to do open-water-style rerouting through ice; safety-

critical, human-in-the-loop decision domain; short season means delays are expensive.

2.5 Scientific and software-engineering challenges

- **Scientific:** chaotic ice / atmosphere dynamics; low observational density in the Southern Ocean; iceberg-forcing uncertainty (size / draft, currents in ice); forecast-skill decay; validation scarcity.
- **Software:** heterogeneous, large, gridded + point + image datasets; geospatial reprojection across multiple polar grids / CRS; asynchronous observation times; data gaps (polar night; clouds for optical; SAR coverage gaps); real-time and offline operation; trustworthy UI; reproducible, auditable pipeline.

2.6 What would make the solution genuinely useful vs. superficially impressive

- **Useful to NCPOR:** a working, honest pipeline on a real route (e.g., Cape Town → Bharati in a chosen season) whose output a marine-logistics officer can sanity-check; a credible MVP; clear method and uncertainty; open data.
- **Impressive-but-scientifically-weak:** a slick dashboard with an unvalidated “AI” that draws a route with no uncertainty, no vessel / ice physics, no forecast coupling, no verification, and claims of “novel AI navigation.” The evaluators (MoES / NCPOR scientists) will discount the latter.

2.7 What the problem statement explicitly requires vs. assumptions / possible extensions

Explicit: an AI / ML-enabled platform that (1) forecasts Antarctic sea-ice concentration, (2) predicts iceberg trajectories, and (3) identifies safe and fuel-efficient navigation routes — using satellite, oceanographic, and meteorological datasets. **Assumed / extension (NOT in the statement; must be argued, not invented):**

- A region focus on NCPOR routes (reasonable but an assumption).
- Real-time / onboard vs. offline planning (the statement is silent — we *propose* a coastal + offline mode).
- Any specific deep-learning architecture, dataset, optimization algorithm, or exact cost function (all engineering choices for us).
- The statement asks for **decision support**; it does *not* ask for autonomous ship control. We should not build “self-driving ship” as the pitch.

3 Real-World Operational Context

The Antarctic sailing season. Antarctic waters are navigable for most research operators only in austral summer (Nov–Mar). Sea-ice extent peaks in ~September (~18 million km² circumpolar) and reaches its minimum in ~February (~3 million km²); the western Antarctic Peninsula is the only area open most years. The NCPOR 39th ISEA advertisement (a primary operational source) confirms the Cape Town → Bharati leg is 10–12 days and the Bharati ↔ Maitri leg 5–7 days “depending on the weather and sea-ice conditions,” Nov–Mar only, using chartered ice-capable ships (with air access via DROMLAN for personnel). India currently has **no dedicated polar ice-breaker**; it charters ice-strengthened vessels for logistics, uses ORV *Sagar Nidhi* / *Sagar Kanya* for oceanography, and is developing a first indigenous ice-class polar research vessel (approved in principle, 2025). The system must therefore handle **varying ice-class vessels**, not one fixed ship. **Why “road-navigation thinking” fails here** (elaborated in Section 12): there are no roads and no static obstacles — ice is drifting and deforming; the “map” changes daily; fuel and speed are strongly nonlinear functions of ice concentration / thickness and ice class; a wrong route can mean days lost or besetment, with search-and-rescue extremely hard; communication latency / limits preclude frequent cloud sync; and human judgement and explainability are paramount be-

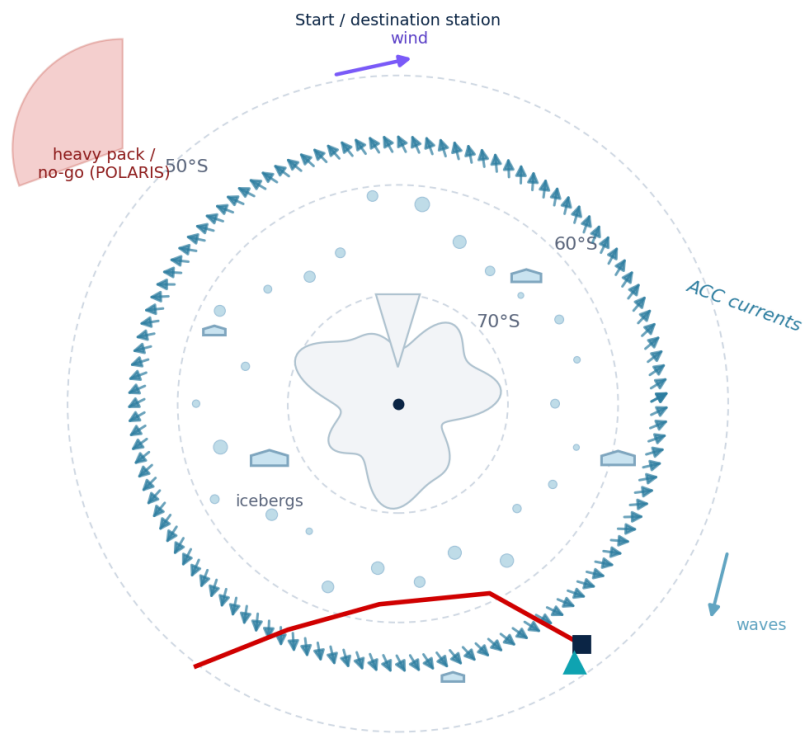


Figure 1: Why Antarctic navigation is difficult (schematic polar view). A research vessel transiting from an open-water approach toward an Antarctic station must thread through a moving marginal ice field that contains drifting icebergs (present year-round), ride the strong Antarctic Circumpolar Current (ACC) ring, and contend with wind, waves and heavy-pack / no-go ice — under sparse data and violent weather. This is a *moving-obstacle, time-varying* environment, not a static road network. (Original schematic illustration; not a measured chart.)

cause lives and expensive vessels are at stake.

4 Scientific Background

4.1 Sea ice, icebergs, ice shelves, glacier ice — the essential distinctions

- **Sea ice** — frozen seawater; forms / ages over weeks–months and is relatively thin (a few cm to several m in Antarctica, often as pack ice, pancake and frazil in the Southern Ocean because of wave action and the open boundary to the north). *It is what a ship must plough through or route around*; it drifts with currents and wind and forms / decays seasonally. Threat: resistance, besetment, hull damage. Managing it is mostly a *concentration / thickness + drift* problem.
- **Icebergs** — large chunks of *glacial* ice that calve from ice shelves / glaciers. They can be massive and tabular (hundreds of km²; A23A and B22A are examples used in drift research) with drafts of hundreds of metres; they drift largely with *ocean currents*, modified by wind, Coriolis and sea-ice drag. Threat: catastrophic collision; they persist year-round, far north of the ice edge. Managing them is a *detection + trajectory prediction* (floating-body dynamics) problem, very different from sea-ice “concentration.”
- **Ice shelves** — floating extensions of grounded glacier ice attached to the continent (e.g., Ross, Filchner–Ronne, Amery). Not directly navigable; they are the calving source of tabular icebergs and a vessel hazard near the grounding / calving front.
- **Fast ice** — sea ice anchored to the coast / shore; it blocks station access (icebreakers “open” McMurdo Sound each year to reach the US station).
- **Polynyas & leads** — openings in the ice; both navigational opportunities and hazards.

Takeaway for a DSS: treat **sea ice as a field** (concentration / thickness → transit cost and risk) and **icebergs as discrete moving hazards** (object-level tracks → exclusion / risk zones). They require different data and different models — precisely why fusing them is non-trivial and non-obvious.

4.2 Why Antarctic navigation is hard

1. Ice is ubiquitous, mobile and heterogeneous; the ice *edge* migrates thousands of km seasonally.
2. The pack is often a mix of open water, pancake / frazil and compacted ice — difficult for SAR and passive-microwave sensing to characterise, and marginal-ice-zone SIC products carry the largest errors.
3. Icebergs are present *everywhere* all year, including far from the ice edge — a “clear-water” route is not iceberg-free.
4. Weather is violent (frequent cyclones, katabatic winds, huge seas); visibility (polar night / cloud) limits optical sensing.
5. In-situ data are sparse: few buoys / current meters; bathymetry and currents near ice shelves are poorly known; models are less well constrained than in the Arctic.

4.3 How sea-ice concentration affects a vessel

SIC (0–100%) plus ice thickness / type / stage control the **resistance** the hull experiences, hence the achievable speed and fuel at a given installed power, and the **risk** of besetment / damage. Operators use schemes such as **POLARIS** (Polar Operational Limit Assessment Risk Indexing System), which map ice regime + ice class to a “Risk Index Outcome (RIO)” that gates whether navigation is permitted at all. A vessel of ice class PC7 cannot transit multi-year ice that a PC3 vessel could; conversely, the fuel / ice-resistant speed curve means the *cheapest / fastest route is rarely the straight line* and is rarely in heavy ice.

4.4 Why safe $\neq$ fuel-efficient

Safety wants the vessel *away* from ice and icebergs (conservative, longer). Fuel efficiency (and schedule) wants the *shortest / least-resistance* route, which may skirt or enter pack ice. These conflict; a DSS must expose the Pareto frontier and let the operator choose — not silently pick one. This is a core reason the system is a **decision-support** trade-off tool.

4.5 Iceberg movement — the driver physics

First-order iceberg drift is a **force balance**: momentum = wind drag + water / current drag (form + skin) + Coriolis + sea-surface pressure-gradient + (in ice) ice-contact / sea-ice drag. To first order, **large tabular icebergs move with the ocean current**, and wind adds a smaller, size-dependent component (the drift-to-wind ratio and turning angle depend on submerged-vs-aerial drag area, i.e., geometry). For a two-dimensional, mostly-advected model one often writes the kinematic relation

$$\mathbf{v}_{\text{berg}} \approx \mathbf{v}_{\text{current}} + \gamma \mathbf{v}_{\text{wind10}}, \quad (1)$$

where γ is an empirical wind-drift factor. Because current and wind are uncertain and ice geometry is only coarsely known, **trajectory uncertainty grows rapidly** — hence ensemble / particle approaches, rather than single-line predictions, are the scientifically honest output (Section 15).

4.6 What a navigation DSS actually does, and the decisions it supports

Not route *autopilot*. It answers, for a human:

- “What will the ice be here in 3 / 7 / 14 days?” (SIC forecast + confidence);
- “Where will tracked icebergs / this iceberg be in T hours–days, and what is the uncertainty cone?”;
- “Which cells are no-go given this vessel’s ice class (POLARIS)?” (navigability map);
- “Among safe-enough routes, which is fastest / lowest-fuel / lowest-risk, and how do they trade off?” (candidate routes);
- “Why is this route suggested, and what happens if the forecast is wrong?” (explainability / sensitivity).

5 Literature Review (Scope and Big Picture)

This is a thematic method survey supporting Sections 6–14. Antarctic-specific vs. Arctic / general evidence is labelled. Full citations are consolidated in Sections 20–21. The literature is best organised around five research themes that map directly onto the system’s components and its integration challenge.

5.1 Research theme A: Antarctic and polar sea-ice forecasting (mature at climate / sub-seasonal scale)

- **ConvLSTM** has become the workhorse. Dong et al. (2024, *Ocean Modelling*) built a ConvLSTM to forecast 60 days of daily Antarctic SIC from SIC alone (1989–2016), skilful to ~ 1 month and catching seasonal extent extremes ~ 1 month ahead. A companion seasonal paper (Dong et al., *Adv. Atmos. Sci.* 2024) showed it beating the ECMWF SEAS5 dynamical system for Antarctic SIC / SIE in a rapid-decline scenario.
- **IceNet** (Andersson et al., *Nature Communications*, 2021) is the reference probabilistic deep-learning seasonal system (trained on CMIP6 + observations; a U-Net ensemble $\rightarrow$ probabilistic SIC maps out to 6 months; beats a dynamical model on seasonal extremes). Its *probabilistic-map* output design is a model for our SIC component — but it is Arctic-first.
- **Community benchmark context: SIPN South** (Massonnet et al., *Front. Mar. Sci.* 2023) coordinates spring-to-summer Antarctic seasonal forecasts; a key finding is that **statistical** /

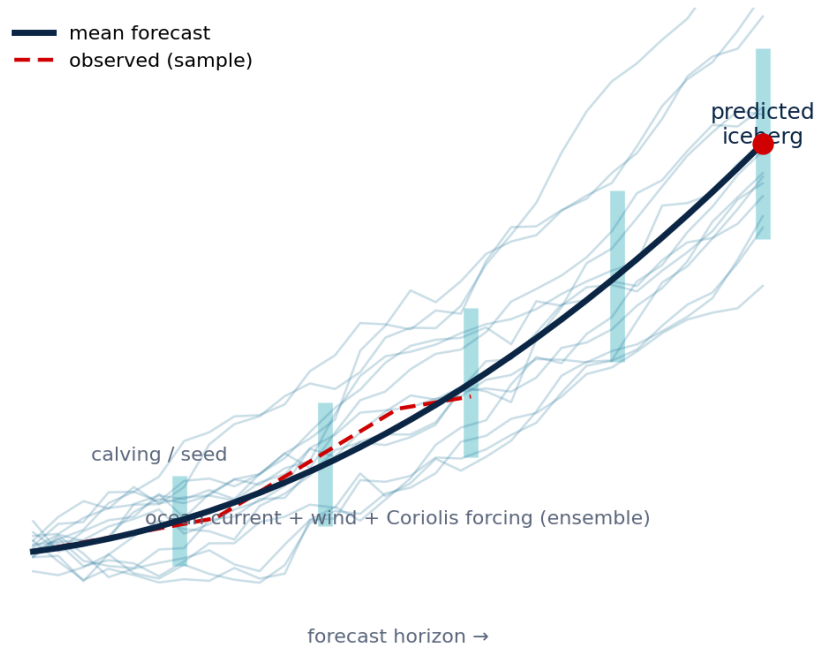


Figure 2: Iceberg trajectory prediction and its uncertainty (schematic). Because the forcing (ocean current, wind, geometry) is uncertain, an ensemble of physically consistent drift realisations fans out from the seed iceberg; the mean forecast and an uncertainty cone are shown, with a sample observed track for context. Reporting this spread — not a single line — is what makes the output useful and honest for routing. (Conceptual illustration.)

ML forecasts generally outperform dynamical coupled models for regional Antarctic SIC / SIE variability. This is strong evidence that an ML SIC forecaster is scientifically credible for Antarctic — for climate / seasonal scales.

- Multi-task and physics-flavoured variants (CNN+ConvLSTM predicting SIC and SIE together; adding EOF / physical constraints to ConvLSTM) show accuracy gains and are relevant as later extensions.
- **Gap at the navigator’s horizon / scale:** the ML Antarctic forecasting literature is overwhelmingly *monthly-to-seasonal and basin-scale*. Few published systems give the *daily, ~1–2-week, sector / route-region* SIC forecast a navigator needs. We can fill this by *re-targeting an established architecture (ConvLSTM / UNet) to a shorter lead and a route corridor* — defensible and feasible, and not a “novel algorithm” claim.

5.2 Research theme B: High-resolution SIC retrieval and monitoring (the “current-state” layer)

- Wulf et al. (2024, *The Cryosphere*) describe **ASIP**: deep learning from Sentinel-1 SAR + AMSR2 passive microwave to produce pan-Arctic, high-resolution SIC with *calibrated uncertainty* ($R^2 \approx 95\%$ vs. regional ice charts). It demonstrates the modern retrieval trend and the importance of uncertainty — but it is Arctic, and retrieval $\neq$ forecast.
- National ice services use such products for tactical navigation; their stated gap is that few add well-calibrated uncertainty and few are operationalised — a pointer for how to present our SIC layer credibly (put uncertainty on every forecast).

5.3 Research theme C: Iceberg detection and tracking (object-level; methods mature, Antarctic-specific work growing)

- **Detection:** Sentinel-1 SAR-based deep learning is now standard. Examples include U-Net / ResUNet segmentation of giant and even sub-km² icebergs (Redmond et al. 2023, *The Cryosphere*;

Jiao et al. 2026, *ESSD* — ResUNet detecting down to 0.016 km², F1 > 0.91, 39,619 grounded icebergs mapped); Inception-v3 classifiers and the YOLO-based **IByolo** on Sentinel-1 wave-mode data (*Remote Sens. Environ.* 2025 / 2026); and semi-automated circum-Antarctic iceberg datasets via random forest on Google Earth Engine (Liu et al. 2026, *ESSD*). The detection / tracking “front end” therefore exists and is Antarctic-tested — **use it rather than claim novelty** — but note it is largely research products, not yet an operational route-safety feed.

- **Trajectory:** physical force-balance drift models are foundational (Bigg et al. 1996; Sodhi & El-Tahan 1980). Antarctic-specific: Wesche & Dierking (2016) developed a wind+current drift model with a sea-ice component, validated on Weddell Sea buoys and used for iceberg re-detection in SAR. Operational-style ML hybrids exist for the Labrador / Arctic offshore context (Yulmetov & Ralph hybrid gradient-boosted trees; Andersson et al. statistical ocean-current VAR model), showing that hybrid physics+data can beat a purely mechanistic model when tuned, at least out to ~24–36 h.
- **Very recent hybrid: IDRIFTNET** (2025, arXiv) combines a physics-driven (Wagner / Bigg-type analytical) drift formulation with residual spectral deep learning, forecasting Antarctic giants A23A and B22A and reporting lower ADE / FDE than pure-DL baselines. It validates the “**physics core + learned residual + ensemble**” direction for icebergs under data scarcity.

5.4 Research theme D: Ice-aware and weather routing (mature in Arctic; Antarctic sparse)

- Foundational framing: ice routing = graph / wave-front optimization over a non-stationary ice field, with a ship-transit model + risk model + cost objective (see e.g. *Optimal ice routing of a ship with icebreaker assistance; A method for ice-aware maritime route optimization*).
- **POLARIS-based routing** (Jeong et al., *Ocean Engineering* 2021; Zhang et al.; Kröger cruise-ship routing) computes an RIO from ice class + ice regime and optimises power / speed with A*/genetic seeds. This is the dominant, standards-grounded way to turn ice into a *risk constraint* — we adopt it.
- **Multi-objective optimisers:** a 3-D ant-colony algorithm (*Ocean Eng.* 2022) for fuel+risk; NSGA-III with A* initialisation for Pareto-optimal polar routes (Wang et al. 2026); an improved D* Lite for dynamic re-routing (2026, Arctic freezing / melting); and A* + rolling-window + line-of-sight for dynamic obstacles (Ma et al., CCDC 2026 — the supplied IEEE reference). These are mostly Arctic / generic but transferable in spirit.
- **Real-time operational tools: RouteView 2.0** (*Environmental Modelling & Software* 2025) for the Arctic Northeast Passage is a rare *deployed* dynamic planner (improved A*, real-time ice / sea-fog / icing). **BAS PolarRoute** (Smith et al., ~2022–24) is the notable *Antarctic* prototype — meshed environment graph + vessel speed / fuel functions + multi-objective route for RRS *Sir David Attenborough*, with a public code repository (github.com/antarctica/PolarRoute). **Study it as the closest existing architecture to what NCPOR needs.**
- Fuel / voyage tools for Arctic transits (e.g., the Europe–Asia voyage-planning tool of Li et al.) combine ice-resistance prediction + fuel minimisation with POLARIS as a constraint.

5.5 Research theme E: Forecast verification and metrics

SIPN / SIPN-South and the forecast community converge on anomaly correlation (ACC), RMSE / MAE, integrated ice-edge error (IIEE), Brier-type probabilistic scores, and skill vs. persistence / climatology baselines. Route evaluation uses time, fuel, risk exposure and robustness (Section 17).

5.6 Research theme F: Operational user-needs evidence

“Sea-ice information and forecast needs for industry maritime stakeholders” (*Int. J. Digital Earth*, 2020) surveys what marine users actually want: routine ice products, near-real-time satellite imagery, short-range met-ocean + ice forecasts, and ice drift / pressure forecasts for continuous rout-

ing. It notes that **Antarctic activity concentrates on the western Peninsula Dec–Apr, and that all Antarctic waters carry iceberg risk year-round** — a strong citation for our product definition.

How the themes connect. SIC forecasting (theme A / B) tells *where the ice field will be*; iceberg detection + drift (theme C) tells *where discrete hazards will be*; ice-aware routing (theme D) turns both into a navigable, least-cost route. The recurring shortfall across all themes is **integration, short-lead / route-region targeting, calibrated uncertainty, and Antarctic operational specificity** — the very space this project occupies.

6 Analysis of the Supplied Reference Papers

All five supplied documents were fetched and examined. Findings are reported transparently; nothing is claimed about any paper that could not be read. **A key conclusion for the team:** only one of the five supplied documents (Ref. 5) is a strong, on-topic scientific reference. Refs. 2–4 are generic or adjacent path-planning papers of limited to moderate relevance and **must not be presented as Antarctic sea-ice / iceberg navigation research**. Ref. 1 could not be accessed and was not cited. The genuine evidence base for the project therefore comes from the verified literature in Sections 5, 7 and 20.

Table 1: Summary assessment of the five supplied reference documents.

Ref	Document	Assessment	Use in this project
1	ScienceDirect (Elsevier) PDF, PII S0029801826030751, <i>Ocean Engineering</i>	Not accessible / unverified. Temporary signed URL expired. Title / authors / DOI not confirmed.	<i>Do not cite.</i> Retrieve persistent DOI before use.
2	Indonesian Journal of Geography — UAV maritime-surveillance path planning, North Natuna Sea	Tangential. UAV (not a vessel), not polar, no ice / iceberg science.	Generic “risk-as-cost-field” idea only; not polar evidence.
3	IEEE CCDC 2026 — improved A* + rolling-window dynamic ship routing	Generic but useful technique. Graph-based ship routing / dynamic re-planning.	Algorithmic reference for the dynamic re-routing module. Not domain evidence.
4	Springer (NT 2026) — mecanum-wheel warehouse AGV, genetic algorithm	Tangential. Warehouse / TSP robot routing.	Essentially nil to polar navigation; do not cite as polar research.
5	<i>The Cryosphere</i> 18:5277 (2024) — pan-Arctic SIC from SAR + passive microwave (Wulf et al.)	High-quality, on-topic. DL SIC retrieval with calibrated uncertainty.	Nowcast / uncertainty philosophy and SAR+PMW fusion rationale; <i>Arctic, retrieval.</i>

6.1 Ref. 1 — ScienceDirect PDF: inaccessible, unverified

The URL is a temporary signed `pdf.sciencedirectassets.com` link that returned a fetch error (expired / invalid). Its PII string S0029801826030751 indicates an *Ocean Engineering* (Elsevier) manuscript, but the exact title, authors, year and DOI could **not** be confirmed from Crossref or the publisher. Per the project’s integrity rules, its content is *not* cited here. Recommendation: retrieve the persistent link / DOI from the ScienceDirect session and re-analyse once available. *Do not cite its content until verified.*

6.2 Ref. 2 — Indonesian Journal of Geography: tangential

A Risk-Based Geospatial Optimization Framework for UAV Maritime Surveillance Path Planning in the North Natuna Sea. UAV (not a ship) maritime-surveillance path planning, North Natuna Sea (Southeast Asia) — neither Antarctic nor polar, and not vessel-routing in ice or ice / iceberg science. Its only relevance is as a generic example of risk-weighted geospatial path planning (multi-objective shortest-path over risk surfaces), an idea that the polar routing literature establishes far better.

6.3 Ref. 3 — IEEE (CCDC 2026): generic routing, moderately relevant

Dynamic Ship Route Planning Based on Fusion of Improved A-Star and Rolling Window Methods. Authors: Kungan Ma, Lipeng Wang, Miao Yu. 38th Chinese Control and Decision Conference, 2026, IEEE. DOI [10.1109/CCDC69976.2026.11560307](https://doi.org/10.1109/CCDC69976.2026.11560307). It improves A* for large environments and dynamic obstacles via (1) sub-region partition, (2) line-of-sight local-target selection, (3) a rotation strategy when the target falls in an obstacle, and (4) local A* with rolling-window-updated dynamic-obstacle information. Relevance: confirms the A*/rolling-window / local-replanning *technique* suited to our dynamic-ice re-routing sub-problem. Limits / do-not-copy: generic obstacles, no drifting-sea-ice or iceberg physics, no POLARIS / ice-class, no Antarctic; no quantitative results in the accessible text.

6.4 Ref. 4 — Springer chapter: tangential (warehouse AGV)

Design and Implementation of a Mecanum-Wheel AGV with Genetic Algorithm-Based Routing Optimization, in *New Technologies, Development and Application IX* (NT 2026), pp. 61–70. DOI [10.1007/978-3-032-30113-0_5](https://doi.org/10.1007/978-3-032-30113-0_5). A ROS2 / SLAM warehouse mobile robot with a Hamming-distance-initialised genetic algorithm solving a TSP. Relevance to the Antarctic problem is essentially nil; do not cite as polar research. Its GA diversity / exploration ideas are at most a minor option if genetic multi-objective routing is used — but the polar-specific A*/NSGA / POLARIS literature serves that purpose better.

6.5 Ref. 5 — Copernicus *The Cryosphere*: highest-quality supplied document (Arctic SIC retrieval)

Pan-Arctic sea ice concentration from SAR and passive microwave. Authors: Tore Wulf, Jørgen Buus-Hinkler, Suman Singha, Hyeon Shi, Matilde Brandt Kreiner. *The Cryosphere* 18, 5277–5300 (2024). DOI [10.5194/tc-18-5277-2024](https://doi.org/10.5194/tc-18-5277-2024). Open access (CC-BY). Objective: an automated, pan-Arctic, all-season, high-resolution SIC retrieval (“ASIP”) from Sentinel-1 SAR + AMSR2 with well-calibrated uncertainty, trained on regional ice charts as labels. Method: deep-learning vision on collocated Sentinel-1 HH / HV + AMSR2 brightness temperatures; $R^2 \approx 95\%$ vs. held-out regional ice charts; generalises pan-Arctic; consistently higher SIC than passive-microwave-only products (biases +1.45 to +8.55%), discrepancies concentrated in the marginal ice zone. Relevance: (a) authoritative example of a *high-resolution, uncertainty-labelled* SIC “current-state” layer; (b) its rationale for fusing SAR (high resolution) with passive microwave (coverage) is directly portable to Antarctic marginal-ice zones; (c) it argues that automated SAR products need calibrated uncertainty before operational / assimilation use — mirroring our uncertainty emphasis. Do-not-copy: it is *retrieval (nowcasting)*, *not forecast*; *Arctic*, not Antarctic; it uses labelled regional ice charts that do not exist operationally everywhere; and it is training / compute-heavy (our prototype should use passive-microwave SIC as the primary field and treat SAR as optional).

7 Recent Research Highlights (2020–2026)

The most decision-relevant recent work, curated from the verified literature. Antarctic-specific is marked [A], Arctic [Ar], general / adjacent [G]. Starred items in Section 20 are the recommended starting points.

- [Ar] **IceNet** — Andersson et al. (2021), *Nat. Commun.* Probabilistic deep-learning sea-ice forecast to 6 months; reference design for probabilistic map outputs.
 - [A] **Antarctic ConvLSTM** — Dong et al. (2024), *Ocean Modelling* 190:102386. 60-day Antarctic SIC, skill ~ 1 month.
 - [A] **Seasonal DL Antarctic** — Dong et al. (2024), *Adv. Atmos. Sci.* ConvLSTM beats ECMWF SEAS5 for Antarctic SIC / SIE seasonal prediction.
 - [A] **SIPN South** — Massonnet et al. (2023), *Front. Mar. Sci.* Community Antarctic seasonal forecast; statistical / ML often beat dynamical models.
 - [Ar] **ASIP high-res SIC** — Wulf et al. (2024), *The Cryosphere*. SAR+PMW SIC retrieval with calibrated uncertainty (supplied Ref. 5).
 - [Ar] **Physical-constraint ConvLSTM** — Liu et al. (2024), *Applied Ocean Research* 148:104007. EOF physical constraints improve short-term (≤ 10 -day) SIC ConvLSTM.
 - [A] **Giant-iceberg segmentation** — Redmond et al. (2023), *The Cryosphere* 17:4675. U-Net maps extent of giant Antarctic icebergs.
 - [A] **Circum-Antarctic grounded icebergs** — Jiao et al. (2026), *ESSD*. ResUNet Sentinel-1 detection to 0.016 km^2 ; 39,619 grounded icebergs (dataset).
 - [A] **Six-year circum-Antarctic iceberg dataset** — Liu et al. (2026), *ESSD* 18:147. Sentinel-1, 2018–2023, per-October, $\geq 0.04 \text{ km}^2$ (dataset).
 - [A] **IByolo / wave-mode icebergs** — *Remote Sens. Environ.* (2025 / 2026). Inception-v3 and YOLOv11 iceberg detection in Sentinel-1 wave mode; $> 80\text{k}$ detections 2016–18.
 - [Ar / G] **IDRIFTNET** — 2025, [arXiv:2507.00036](https://arxiv.org/abs/2507.00036). Hybrid physics + spectral-DL iceberg drift forecast (A23A / B22A).
 - [Ar] **Hybrid ML iceberg drift (operational)** — Yulmetov & Ralph (2019 / 2021, POAC / C-CORE). Physics + gradient-boosted-trees tactical iceberg drift forecast, Labrador.
 - [Ar] **Statistical ocean-current iceberg forecast** — Andersson, Scibilia, Imsland (2019), *Cold Reg. Sci. Technol.* VAR current model + iceberg drift; $\lesssim 36$ h.
 - [A] **Wind / current Antarctic iceberg drift** — Wesche & Dierking (2016), *Cold Reg. Sci. Technol.* Weddell Sea drift model + sea-ice component + SAR re-detection.
 - [Ar] **POLARIS ship route planning** — Jeong et al. (2021), *Ocean Engineering* 234. RIO risk + speed / power optimization.
 - [Ar] **3-D ant-colony ice routing** — (2022), *Ocean Engineering*. Fuel + risk multi-objective ice routing.
 - [Ar] **NSGA-III polar multi-objective** — Wang et al. (2026), *Ocean Eng.* Pareto route planning (distance / time / fuel / risk).
 - [Ar] **Improved D* Lite dynamic routing** — (2026). Dynamic re-planning in freezing / melting Arctic.
 - [Ar] **RouteView 2.0** — (2025), *Environmental Modelling & Software*. Deployed real-time Arctic route planner.
 - [A] **PolarRoute / Autonomous Passage Planning for a Polar Vessel** — Smith et al. (BAS, ~ 2022 –24). Antarctic multi-objective route planning on an unstructured mesh for RRS *Sir David Attenborough*; github.com/antarctica/PolarRoute.
 - [G] **Multi-objective ice routing with icebreaker assistance and ice-aware maritime route optimization** — foundational two / three-tier routing-cost framing.
 - [Ar / A] **Ice-class / ice-regime standards** — IMO Polar Code; POLARIS risk indexing (ABS / IACS).
- Old-but-foundational (still load-bearing):** Bigg et al. (1996) iceberg-trajectory prediction; Sodhi & El-Tahan (1980) iceberg drift during storms (force balance); Shi et al. (2015) ConvLSTM (origin of the standard model).

8 Research Comparison Table

The table below consolidates the strongest, verified works relevant to the three components and to the Antarctic context. Labels: **A** = Antarctic evidence, **Ar** = Arctic evidence, **G** = general / adjacent. The final column reflects suitability for an SIH *prototype* vs. real *operational* deployment. Reported results are those stated by each paper’s authors and are not claimed for our (unbuilt) system.

Table 2: Comparison of the most relevant research papers (the same works consolidated in Section 20). Region labels: **A** = Antarctic, **Ar** = Arctic, **G** = general / adjacent. Reported results are as stated by each paper’s authors.

Paper (year, region)	Problem	Method / dataset	Reported results	Limitations	Relevance / suitability
Dong et al., <i>Ocean Modelling</i> (2024) [A]	Antarctic SIC / SIE forecast	ConvLSTM; NSIDC SIC 1989–2016	Skilful ~1 mo daily SIC; SIE monthly error mostly <0.2 M km ² ; seasonal extrema ~1 mo ahead	SIC-only input; basin scale; regime-shift sensitivity	SIC forecast backbone. Prototype: high (SIC open).
Dong et al., <i>Adv. Atmos. Sci.</i> (2024) [A]	Antarctic SIE / SIC seasonal	ConvLSTM re-forecast Dec–Jun; NSIDC SIC	Beats ECMWF SEAS5	Seasonal, basin scale	DL credibility for seasonal; not route-scale.
Massonnet et al., SIPN South (2023) [A]	Coordinated Antarctic seasonal forecast	Community intercomparison (22 contributors, >3000 forecasts)	Statistical / ML generally beat dynamical models regionally; group median captures recent lows	Summer-season focus; regional skill gaps (Ross)	Justifies an ML SIC approach to NCPOR. High.
Andersson et al., IceNet (2021) [Ar]	Probabilistic DL SIC forecast ≤6 mo	U-Net ensemble; CMIP6 + obs	Beats a dynamical model on seasonal extremes	Arctic-first; coarse	Template for probabilistic SIC output.
Wulf et al., ASIP, <i>TC</i> (2024) [Ar]	High-res SIC retrieval w/ uncertainty	DL vision; Sentinel-1 + AMSR2 + ice charts	$R^2 \approx 95\%$ vs charts; MIZ biases +1.5–8.5%	Retrieval≠forecast; Arctic; scarce labels	Nowcast / uncertainty philosophy; SAR fusion.
Liu et al., <i>Appl. Ocean Res.</i> (2024) [Ar]	Short-term (≤10 d) SIC forecast	ConvLSTM + EOF physics constraints; re-analysis + SIC	EOF constraints improve SIC prediction	Arctic; ≤10 d	Physics-in-DL template. Prototype: medium (later).
Redmond et al., <i>TC</i> (2023) [A]	Giant-iceberg extent segmentation	U-Net; Sentinel-1	Faster / more robust than Otsu/ k -means; outlines giant icebergs	Giant only; manual init	Detection prior. Prototype: med-high (SAR + labels).
Jiao et al., <i>ESSD</i> (2026) [A]	Grounded-iceberg detection / ID	ResUNet + multi-temporal matching; Sentinel-1 SAR + IBCSO + SIC	F1 > 0.91; min size 0.016 km ² ; 39,619 icebergs	Grounded (stationary) only; data product	Detection / ID prior; use precomputed catalogues.
Liu et al., <i>ESSD</i> (2026) [A]	Circum-Antarctic iceberg dataset	Incremental random forest + manual; Sentinel-1 (GEE) 2018–23	Per-October icebergs ≥0.04 km ²	Annual snapshot (Oct); semi-auto	Observation catalogue for validation / tracks. High.
IByolo / wave-mode (2025 / 26) [A]	Iceberg detection in open ocean	YOLOv11 / Inception-v3; Sentinel-1 WV SAR	AP50:95 0.41; 80k+ detections 2016–18	WV vignettes only (small chips)	Open-ocean detection; near-real-time potential.

continued on next page

Table 2 (continued).

Paper (year, region)	Problem	Method / dataset	Reported results	Limitations	Relevance / suitability
IDRIFTNET (2025) [A]	Iceberg drift forecast	Physics drift + residual spectral DL; A23A / B22A + ERA5 / current	Lower ADE / FDE than Trajectron++, GroupNet, PECNet	Two icebergs; not yet peer-reviewed; scarce data	Best current template for hybrid iceberg. High.
Yulmetov & Ralph (2019 / 21) [Ar / G]	Tactical iceberg drift (offshore)	Physics core + gradient-boosted trees; GPS + wind / current forecast	Improves drift forecast over mechanistic, $\leq 24\text{--}36$ h	Arctic offshore context	Physics+ML pattern; uncertainty framing.
Andersson et al., <i>Cold Reg. Sci. Technol.</i> (2019) [Ar / G]	Short-term iceberg forecast	Statistical VAR on ocean current	Superior to dynamic model short-term	~ 36 h; Arctic	Simple statistical baseline, short-lead.
Wesche & Dierking (2016) [A]	Antarctic iceberg drift + re-detect	Wind / current drift + sea-ice component; Weddell buoys + wind forecast	Reproduces Weddell drift; enables SAR re-detection	Buoy scale; region-specific	Antarctic drift physics + SAR re-ID. High.
Jeong et al., <i>Ocean Eng.</i> (2021) [Ar]	Polar route + speed optimization	POLARIS RIO + transit model + seed GA (A* seed); Arctic ice + ship	Optimal power / speed for R / V Araon	Arctic; single-objective power	Standards-grounded risk→route. High (adopt POLARIS).
3-D ant-colony, <i>Ocean Eng.</i> (2022) [Ar]	Multi-objective ice routing	3D-ACA (fuel + risk); NSR ice / weather	Fuel / risk Pareto over time-varying ice	Arctic; heuristic tuning	Multi-objective routing reference.
Wang et al., <i>Ocean Eng.</i> (NSGA-III, 2026) [Ar]	Pareto polar routing	NSGA-III + A* init.; modelled env. fields	Pareto-optimal set over four objectives	Arctic; compute-heavy	Multi-objective method; explains trade-offs.
Improved D* Lite (2026) [Ar]	Dynamic re-routing in ice	Improved D* Lite incremental; synthetic freezing / melting	Low update cost vs D* Lite / global	Arctic; simulation	Re-planning when the ice forecast updates. High for replan.
RouteView 2.0 (2025) [Ar]	Deployed real-time Arctic routing	Improved A*, rolling; NEP env. + ship	Real-time operational route planning	Arctic NEP	Rare deployed analogue; functional target.
PolarRoute / Smith et al. (2022–24) [A]	Antarctic multi-objective route planning	Unstructured mesh + path + smoothing; vessel speed / fuel fns	Weddell Sea RRS SDA routes; season-dependent	Research prototype; needs env. forecast feed	Closest existing architecture; emulate and extend. High.
Bigg et al. (1996) [Ar / G]	Iceberg-trajectory prediction	Coupled ocean + iceberg model; North Atlantic / Arctic	Large-iceberg trajectory modelling	Coarse; old	Foundational iceberg-dynamics reference.
Sodhi & El-Tahan (1980) [G]	Iceberg drift during storm	Force-balance model; field radar / current / wind	Good prediction with medium-tabular assumptions	Data-hungry; single case	Physics core of the iceberg model.

8.1 What the comparison table tells us

- **Solved / practical:** SIC forecasting by deep learning (basin / seasonal), iceberg *detection*, single- and multi-objective *ice-routing* algorithms, and POLARIS-based risk. All are feasible for a prototype with open data.
- **Still hard / expensive:** (i) route-region daily Antarctic SIC at a 1–2-week lead with vali-

dated skill; (ii) iceberg trajectories with *reliable* uncertainty; (iii) fusing forecast + iceberg + routing; and (iv) real-time operation under communications limits.

- **Prototype-feasible:** ConvLSTM / UNet SIC (passive-microwave SIC is free); physics + ensemble iceberg drift using existing iceberg catalogues + ERA5 currents; A*/Dijkstra + POLARIS-like routing on reanalysis fields for a chosen season and route.
- **Operational-deployment-feasible:** only with vessel-specific transit curves, near-real-time feeds and validation; the research systems (IceNet, ASIP, RouteView, PolarRoute) show it is possible but heavy.

9 Research Gaps

9.1 The honest starting point: what already exists well

Single-iceberg *detection* (SAR + deep learning) is solved to a research grade; SIC forecasting at climate / seasonal scales is solved to a useful grade; ice-routing algorithms are plentiful and standards-backed (POLARIS); and multiple real tools exist (RouteView, PolarRoute). **These should not be claimed as our novelty.**

9.2 The genuine gaps — and where they come from

1. **No integrated Antarctic forecast→risk→route platform.** SIC forecasting (climate journals), iceberg trajectory (cryosphere / oceanography) and ice routing (naval / ocean-engineering) are published in *different literatures* that almost never cite each other. The supplied references themselves are fragmented (SAR retrieval $\neq$ routing $\neq$ iceberg). A single coherent pipeline for one operator and route is not an off-the-shelf product.
2. **Component mismatch in forecast horizon and scale.** The ML Antarctic SIC work is basin-scale / monthly-seasonal; the route planner needs *route-corridor*, *daily*, ~ 1 – 2 -week SIC with uncertainty. Bridging the two (fine spatial, short lead, probabilistic) is an open, operationally motivated problem.
3. **Iceberg prediction rarely carries honest uncertainty into routing.** Research publishes point trajectories or (rarely) ensemble spread; operators need the *probability of a given iceberg transiting a given route cell in a time window* so the router can add a dynamic exclusion / risk cost. Little exists.
4. **Antarctic (vs. Arctic) routing under ice + icebergs is underrepresented.** The dominant routing literature is Arctic (Northern Sea Route / coastal). Antarctica has fewer validated tools, fewer in-situ / forecast constraints, an open ice edge (more mobile pack, more bergy bits) and chartered heterogeneous ice-class vessels. Extending validated Arctic methods to Antarctic specifics — and to NCPOR’s exact Cape Town–Maitri–Bharati legs — is a real, useful gap.
5. **Vessel-specific ice-transit and fuel data.** Most routing uses generic ship models; a practical Antarctic tool must ingest a given vessel’s ice-class / POLARIS profile and resistance / speed curves — and such profiles for research vessels are rarely public. This is a data gap, not an algorithm gap.
6. **Uncertainty and verification in an operational setting.** Almost every paper flags the lack of calibrated uncertainty and of real-world route verification as open problems. A prototype that *explicitly* reports uncertainty and compares its routes / forecasts against a real archived voyage (even a reanalysis “hindsailed” route) is more defensible than most demo dashboards.
7. **Operational constraints (communications, human-in-the-loop, data latency) are ignored** in most research, yet they are the whole problem for a real operator.

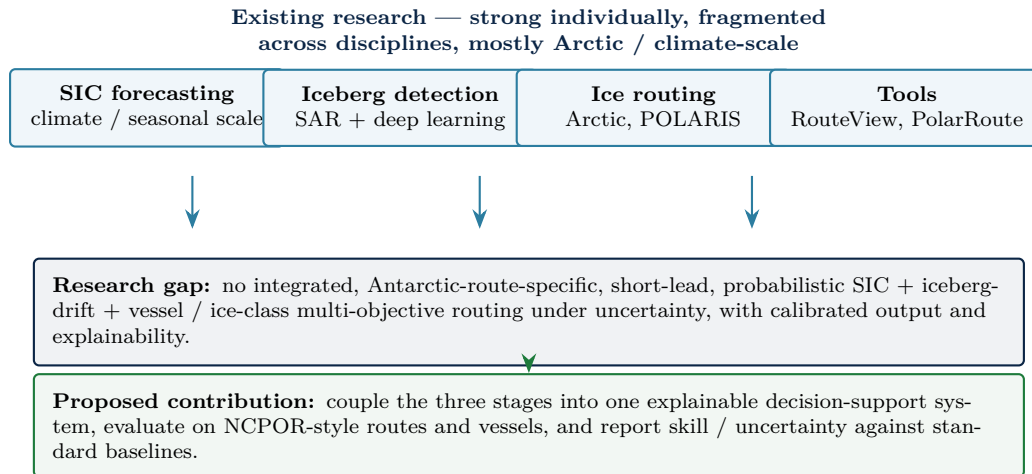


Figure 3: Research-gap framework. Strong but fragmented existing capabilities (top) do not deliver the integrated Antarctic, short-lead, probabilistic, vessel-aware decision-support pipeline (bottom). The project targets exactly this gap.

Conclusion. A credible gap is an **integration + operational + validation gap**: “Coupling a short-lead, route-region, probabilistic Antarctic sea-ice forecast with iceberg-drift prediction and vessel / ice-aware multi-objective routing — evaluated on real NCPOR-style routes and an explicit vessel / ice-class, with calibrated uncertainty and explainable output — is not provided by any existing, accessible system.” That is defensible; claim exactly that.

10 Proposed System Concept

10.1 Inputs

- **Real-time (feasible for an operational prototype):** OSI-SAF / NSIDC / AMSR2 SIC analysis; ERA5 / ECMWF forecast wind and waves; a sea-ice *drift forecast* product if available; current / best-known iceberg positions (from catalogues / ice services if accessible); vessel AIS position and speed.
- **Historical (for training / validation):** multi-decade SIC CDR; ERA5 reanalysis (wind, waves, temperature); ORAS5 / ECCO ocean currents; IBCSO bathymetry; archived iceberg catalogues / tracks; vessel voyage logs if provided.
- **Forecast (the model output, also an input to routing):** the system’s own SIC and iceberg forecasts.
- **Static:** bathymetry (IBCSO), coastlines / land masks, vessel ice-class / POLARIS table, speed-power curves.
- **Optional:** Sentinel-1 SAR (high-resolution SIC / icebergs — heavy; treat as a validation / nowcast layer, not mandatory for the MVP); ocean-colour / cloud-free optical when available (summer only).

Explicit data-tiering honesty: we do *not* assume a “real-time SAR iceberg feed” is available. For the prototype, icebergs are handled via *published Antarctic iceberg catalogues / tracks* (Jiao 2026; Liu 2026; BYU / MERGED if accessible) plus a detection method demonstrated on Sentinel-1 as an optional nowcast.

10.2 Processing

1. **Ingestion & harmonisation:** pull each source onto one Antarctic polar-stereographic (or route-corridor) grid; align temporally.
2. **Cleaning / quality control:** mask land; remove bad pixels (polar-night / optical gaps, satel-

lite coverage); flag missing data.

3. **Feature engineering:** SIC $\rightarrow$ local ice-state descriptors (concentration, drift tendency); a vessel transit-cost function of (SIC / thickness, ice class); risk indices (POLARIS-like RIO); iceberg proximity / dynamic-obstacle fields.
4. **SIC forecasting:** a trained spatiotemporal model $\rightarrow$ daily probabilistic SIC over the corridor to ~ 1 – 2 weeks; a persist / trend baseline for skill comparison.
5. **Iceberg module:** track catalogued icebergs forward with physics drift + ensemble spread $\rightarrow$ time-indexed probabilistic encounter zones.
6. **Risk mapping:** combine no-go (POLARIS / land / safety), ice-transit penalty, iceberg-encounter probability, and weather penalty into cell-cost time layers.
7. **Routing:** time-varying least-cost multi-objective graph search (safe vs. efficient vs. balanced), producing candidate routes + trade-offs; re-run when forecasts update (replanning).
8. **Output & explanation:** forecasts, uncertainty, risk maps, routes, ETA / fuel, per-segment rationale, alerts, what-if sensitivity.

10.3 Outputs (products)

- Daily SIC forecast maps + confidence (edge-of-ice band highlighted).
- Iceberg trajectory set + uncertainty cone, and encounter alerts.
- A navigability map (“green / amber / red”) for the given vessel / ice class.
- A recommended route + 2–3 alternates on a risk / efficiency / fuel trade-off, with ETA and fuel estimate.
- Explainable annotations (why each segment is recommended; what degrades if the forecast changes).
- Logging for verification.

11 Data Sources (verified and accessible)

Prioritised. “Student prototype” = usable with free downloads / API; “Operational NRT” = near-real-time availability (still requires engineering). All datasets below are real, open and documented; none are assumed to be real-time where they are not.

Sea-ice concentration.

- **OSI-SAF Global Sea Ice Concentration CDR (OSI-450 / 430)** (EUMETSAT / OSI SAF, via Copernicus Marine / C3S): daily, 25 km (12.5 km in some products), Arctic + Antarctic, 1978–present; open. Prototype yes; NRT yes. Key SIC reference for fields / training.
- **NSIDC Sea Ice Concentrations, Nimbus-7 SMMR & DMSP SSM / I-SSMIS** (NASA Team / Bootstrap, NSIDC-0051): daily, 25 km, 1978–present.
- **AMSR2 / AMSR-E SIC** (NSIDC-0081 / NSIDC-0803; U. Bremen ASI; NOAA CoastWatch AMSR2 ~ 10 km NRT): higher-resolution (~ 3 – 12 km) passive microwave, good for a high-resolution “current” layer; AMSR-E 2002–2011, AMSR2 2012–present; NRT via NOAA CoastWatch (latency ~ 12 – 24 h). Note that SIC accuracy is lower in marginal ice / melt.
- NSIDC CDR / NOAA sea-ice products for consistency.

Atmosphere and waves (reanalysis and forecast).

- **ERA5 / ERA5T** (ECMWF, Copernicus Climate Data Store): 10-m wind (u / v), waves (significant wave height, period, direction), 2-m temperature, sea-level pressure; ~ 31 km, hourly, 1940–present; free; forecast fields via ECMWF / C3S where licensing allows. Reanalysis is ideal for training / validation. Caveat: studies find ERA5 *underestimates extreme wave height* — a safety flag (routing must not under-estimate extreme waves).

- (Optional) ECMWF / Copernicus operational atmospheric and wave forecasts if accessible (licensing may be a concern; reanalysis is the safe prototype default).

Ocean currents and dynamics.

- **ORAS5** (ECMWF ocean reanalysis): ocean / sea-ice state, $\sim 1^\circ$, 1979–present; free via CMEMS / ECMWF; a standard Southern-Ocean drift forcing (Zuo et al. 2019).
- **ECCO** (NASA / JPL) v4r4: state-estimate ocean; free.
- (Optional, higher-resolution / derived) Copernicus Marine Service GLORYS12 / Mercator Southern-Ocean products. Currents *under* ice remain uncertain — the iceberg module should treat the current as an uncertain forcing (ensembles).
- **Ice drift**: sea-ice-drift products (OSI-SAF / NSIDC) inform how the SIC “current state” moves and iceberg-in-ice drag.

Bathymetry (static).

- **IBCSO v2** (International Bathymetric Chart of the Southern Ocean, version 2; *Sci. Data* 2022): regionally best ~ 500 m, continental-shelf-focused, free. Use for grounding / coast / route feasibility (e.g., staying off shallow shelf / grounded-iceberg zones).

Icebergs (observation catalogues).

- **Circum-Antarctic grounded-iceberg dataset** (Jiao et al. 2026, *ESSD*, IMAS metadata) — Sentinel-1, near-coast, grounded; 39,619 icebergs.
- **Six-year circum-Antarctic iceberg dataset 2018–2023** (Liu et al. 2026, *ESSD* 18:147) — per-October, ≥ 0.04 km², area / location / mass + uncertainty; usable as track seeds (annual, so coarse in time).
- **BYU Antarctic Iceberg Tracking Database** (Budge & Long; “MERGED”/BYU) — long-term giant-iceberg tracks; good for validating the drift module.
- **NIC (US National Ice Center)** — operational positions of named tabular icebergs; near-real-time but sparse / irregular.
- **Sentinel-1 SAR imagery** (Copernicus, free) — raw source for detection / nowcast (EW / HH recommended); download and processing are heavy.

Vessel / route reference.

- **IMO Polar Code** and **POLARIS risk indexing (IACS / ABS)** tables — standards for ice-class→risk; a static lookup.
- Vessel speed–power / ice-resistance curves: ship-specific; for the prototype use published generic research-vessel curves (e.g., those used by KARS / Jeong 2021 or PolarRoute’s functions for the *Sir David Attenborough*) clearly labelled as placeholders.
- Verification / reference: SIPN-South forecast and verification data (public).

12 Real-World Operational Constraints (Why Ordinary Routing Fails)

- **Only-Antarctic-summer navigation**; short fixed windows; delays are not recoverable once the season closes.
- **Ice-class physics**: a ship’s permissible ice regime and speed-in-ice are governed by hull class (POLARIS / IACS). A route safe for a PC3 icebreaker may be impassable for a chartered PC7 supply vessel. The “cost surface” is *vessel-specific*, not a universal road map.

- **Ice is moving:** there are no static obstacles; a “clear” corridor can close in hours; concentration and drift must be forecast and routes re-planned (D* Lite / rolling-window replanning, plus uncertainty).
- **Icebergs everywhere, including open water:** discrete, detail-unpredictable hazards; even open-water routing needs encounter avoidance.
- **Forecast / data uncertainty:** the Southern Ocean is weakly observed; SIC near the ice edge and extreme waves are the least accurate; currents under ice are poorly known, so ice and iceberg forecasts carry large, growing error. Honest uncertainty is mandatory.
- **Communications / connectivity:** high-latitude satellite links are intermittent and low-bandwidth; an **onboard system must run offline / degraded and only sync when a link exists** — not a cloud-requiring real-time app. Design for on-ship deployment with a land-based pre-voyage planning mode.
- **Safety-critical, human-in-the-loop:** a wrong “optimal” can be fatal or expensive; **explainability and operator override are features, not afterthoughts**. No autonomous-control claims.
- **Fuel and time nonlinearity:** speed–power in ice is strongly nonlinear; “shortest” is rarely cheapest; the POLARIS constraint makes many straight lines illegal.
- **Sparse search-and-rescue / emergency limits:** conservatism in risk tolerance is therefore appropriate.
- **Crew fatigue / decision load:** the DSS should reduce, not add, cognitive load — clear products, few and understandable.

13 AI / ML Approach Comparison (Per Component)

Recommendations here are *analysis*, grounded in the cited evidence. Deep learning is recommended only where it clearly earns its place; simpler or physics-based methods are preferred where data are scarce or explainability and safety dominate.

13.1 Component-level recommendations (analysis)

- **SIC:** ConvLSTM as the primary trained component (strongest evidence-to-effort for Antarctic), with a persistence / climatology baseline reported for skill, output as probabilistic / ensemble fields; later optionally a UNet-IceNet-style probabilistic net and physical constraints. Do *not* attempt a coupled numerical model.
- **Iceberg:** physics force-balance core + ensemble / particle uncertainty, optionally with a learned residual later. Use public catalogue tracks (A23A / B22A, circum-Antarctic datasets) for validation. Do *not* use pure-DL sequence models — data scarcity makes them the weakest choice here, and the newest Antarctic evidence (IDRIFTNET) shows physics + residual wins.
- **Routing:** grid / A* or Dijkstra for a primary, time-varying combined-cost route (POLARIS-based risk as a hard constraint), plus a multi-objective (NSGA-III or weighting) layer exposing a safe / efficient / balanced Pareto set, and D* Lite / rolling-window for replanning when the SIC / iceberg forecast updates. *Avoid reinforcement learning* — high effort, low explainability, unnecessary for the MVP, and a weak safety story.

14 Recommended Technical Architecture

Philosophy: hybrid physics + targeted deep learning + standard optimization, wrapped in an explainable decision-support layer. Deep learning is used only where it clearly earns its place (SIC forecasting); physics is used where data are scarce (icebergs); and established graph optimization is used for routing (safety-critical, explainable).

Table 3: AI / ML approaches for sea-ice concentration (SIC) forecasting. Comparison of candidate methods for the forecast component; selection rationale is in the text (Section 14).

Approach	How it works	Strengths	Weaknesses	Data / compute	Explain.	Prototype	Operational
Persistence / climatology (baseline)	SIC stays constant / seasonal mean	Simple; always available	No skill at longer leads near the edge	SIC only; trivial	Yes	Must-report baseline	Baseline
Statistical / AR on SIC	Linear time-series per pixel	Cheap; robust	Weak spatial / spatiotemporal skill	SIC; low	Yes	Yes (baseline)	Yes
ConvLSTM	Convolutional + recurrent across space / time	Proven for SIC (Dong 2024); handles 2-D evolution	Needs SIC time series; errors accumulate	Daily SIC + optional forcing; moderate GPU	Low	Yes (primary)	Yes
UNet-style CNN (IceNet-like)	Encoder-decoder predicts SIC maps	Good spatial structure; ensemble-probabilistic	Design effort; needs training data for long leads	SIC + forcings; moderate-high	Low	Yes	Yes
Transformers (Conv-Transformer hybrids)	Attention over space / time	Strong long-range; seasonal encoding	More data / compute; newer	SIC + season; high	Low	Stretch	Later
Physics-informed (EOF constraints / bias correction)	Impose or correct physical modes	More consistent; helps small data	Added complexity	SIC + reanalysis; moderate	Medium	Later (nice)	Yes (assimilation)
Dynamical (numerical) model	Physics ocean / sea-ice model	Mechanistic	Huge compute; often <i>less</i> skill than ML for Antarctic regional SIC (SIPN South)	Full forcing; very high	Medium (process)	No (out of scope)	Done by NCPOR / ECMWF

Table 4: AI / ML approaches for iceberg trajectory prediction. Comparison for the drift component.

Approach	How it works	Strengths	Weaknesses	Data / compute	Explain.	Prototype	Operational
Kinematic advection (dead reckoning)	$\mathbf{v} \approx \mathbf{v}_{\text{cur}} + \gamma \mathbf{v}_{\text{wind}}$	Trivial; good for hours-days	Needs current / wind; no dynamics	Current + wind; low	High	Yes (baseline)	Baseline
Physics force-balance drift (Bigg / Sodhi / Wagner)	Momentum balance: wind + water + Coriolis	Physically grounded; Antarctic literature (Wesche 2016)	Needs size / draft, currents; calibration	Size + current + wind; low	High	Yes (core)	Yes
Ensemble / particle filter (track + forecast)	Propagate samples with forcing uncertainty	Honest uncertainty / encounter probability	Needs an uncertainty model of the forcing	Current / wind uncertainty; moderate	High	Yes (with physics core)	Yes
Kalman / EKF / unscented filter	Fuse observations + model for state	Good short-horizon track assimilation	Linear / Gaussian limits	Observed tracks; low	Medium	Yes	Yes
LSTM / GRU trajectory net	Learn sequence $\rightarrow$ next positions	Flexible; needs no physics	Data-hungry; weak uncertainty; overfits few icebergs	Many tracks; moderate	Low	No (data scarce)	No
Graph / Transformer trajectory nets	Social / agent trajectory prediction	SOTA for pedestrians / vehicles	Mis-targeted for icebergs; many tracks; pure-DL shown worse (IDRIFTNET)	Many tracks; high	Low	No	No
Hybrid physics + ML (IDRIFTNET, Yulmetov)	Physics core + learned residual / ensemble	Best of both under data scarcity (beats pure-DL on A23A / B22A)	Implementation effort	Tracks + forcing; moderate	Medium	Yes (target)	Yes

Table 5: Approaches for **route planning** in ice-covered polar waters. Comparison for the optimizer component.

Approach	How it works	Strengths	Weaknesses	Data / compute	Explain.	Prototype	Operational
Dijkstra	Uniform-graph shortest path	Exact; simple	Slow on large grids; single-objective; non-adaptive	Static cost; low-med	Yes	Yes (baseline)	Baseline
A* (and variants: rolling-window, LOS, HFS-A*)	Heuristic search	Fast; the dominant ice-routing tool (POLARIS routes; RouteView)	Heuristic / quality trade-off; single-objective	Static + dynamic cost; low-med	Yes	Yes (primary)	Yes
D* Lite / incremental	Re-plan only changed parts	Ideal when the SIC forecast updates / ice closes	Added complexity	Dynamic cost; med	Medium	Later	Yes (re-plan)
Wave-front / iso-cost	Continuous wave propagation	Good for non-homogeneous fields (PolarRoute uses meshes + paths)	Design complexity	Env. + vessel functions; med	Medium	Later	Yes
Genetic / NSGA-III (Pareto)	Evolve a route population; multi-objective	Explicit Pareto safe-vs-fuel-vs-time trade-offs	No optimality guarantee; compute	Env. + vessel; med-high	Medium (Pareto)	Yes (multi-obj layer)	Yes
Ant colony (3D-ACA)	Colony over time-varying ice	Handles fuel + risk in time-varying ice	Heuristic	Env.; med	Medium	Later	Yes
Reinforcement learning	Agent learns a policy	Adaptive to stochastic ice	Needs simulator / rollout; not justified for SIH budget; poor explainability / safety	Large; high	Low	No	No (risk)

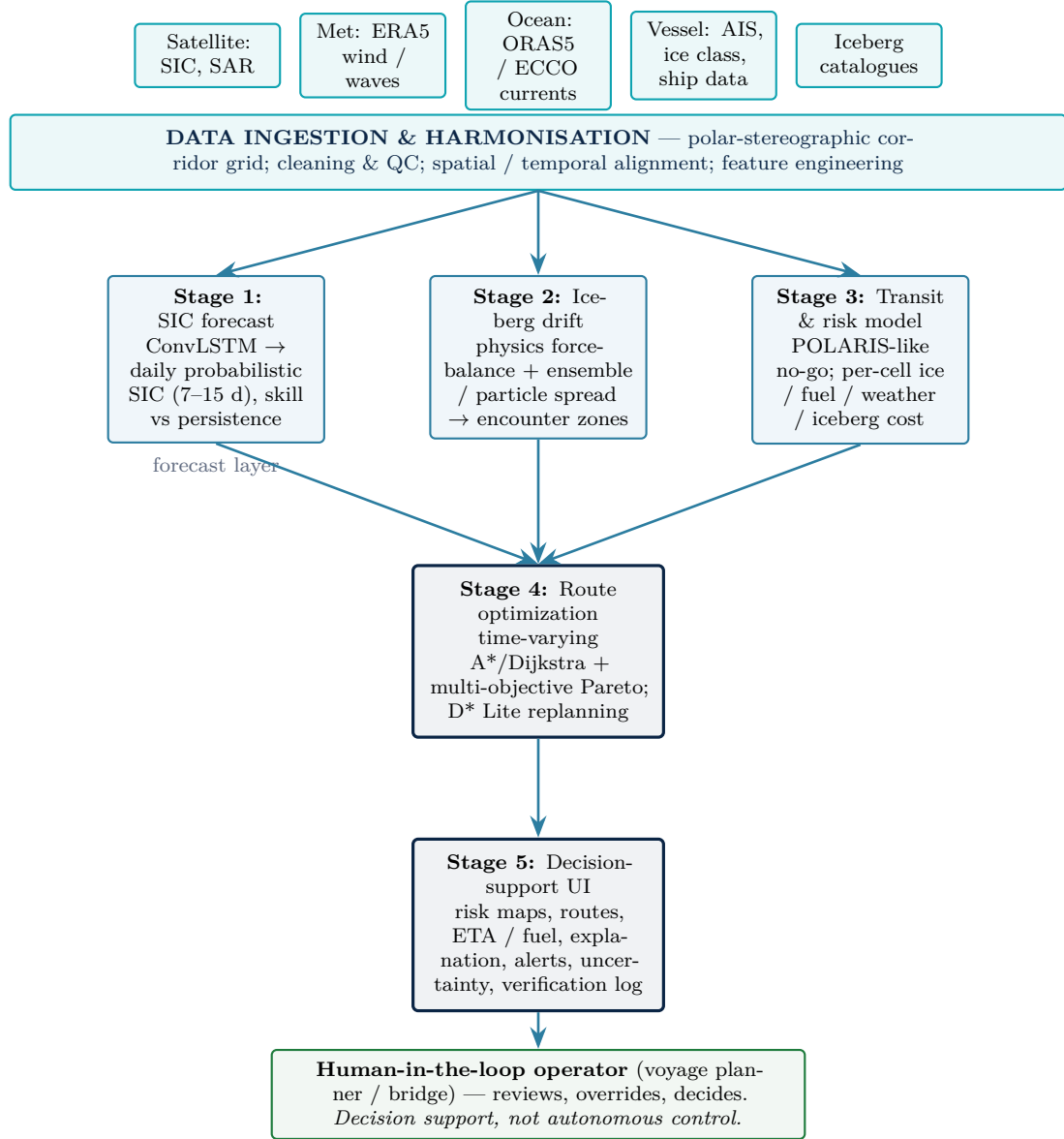


Figure 4: Complete system architecture. Open data are harmonised on a corridor grid and flow through three decoupled scientific stages (SIC forecast, iceberg drift, transit / risk model) into a route optimizer and finally an explainable, human-in-the-loop decision-support interface. Replanning (Stage 4) is triggered when the Stage 1 / 2 forecasts update.

14.1 Core flow (pipeline, decoupled stages, testable individually)

1. **Data layer** (open datasets, Section 11). Harmonise onto an Antarctic polar-stereographic corridor grid; separate real-time / observed vs. reanalysis / historical.
2. **Stage 1 — SIC forecast.** Train a ConvLSTM on daily SIC CDR (optionally + ERA5 / ORAS5 channels) for the corridor / season; produce daily ensemble / probabilistic SIC fields out to $D = 7/15$ days. **Always report skill vs. persistence.** (This is the deep-learning core.)
3. **Stage 2 — Iceberg drift.** From catalogue / seeded icebergs, run a physics drift model with current + wind forcing; add ensemble / particle spread on current / wind / geometry → probabilistic encounter zones over the same D -day window. (Physics + uncertainty; ML residual optional.)
4. **Stage 3 — Transit and risk model.** For the given vessel (ice class / POLARIS table): compute (a) a no-go mask (POLARIS RIO limit, land, shallow, safety buffers), (b) per-cell transit cost = speed / fuel penalty as a function of (SIC, thickness, ice class), and (c) a dynamic hazard cost from the Stage-2 encounter probability + weather (extreme-wave threshold, sea state). Time-index each cost layer over the voyage window.
5. **Stage 4 — Route optimization.** Time-varying A*/Dijkstra least-cost path within the no-go-masked grid; then a multi-objective pass → recommended route + alternates; expose ETA, fuel and risk score. Re-plan with Stage 1 / 2 updates (D* Lite-style incremental).
6. **Stage 5 — Decision-support UI + explanation.** Maps, uncertainty bands, per-segment “why,” alerts, sensitivity (“if the ice arrives 3 days earlier...”), and logs for verification. Human override.

14.2 Build order (what first vs. later)

- **Build first (MVP):** Stages 3–5 on *reanalysis* / *known* SIC + a catalogued static iceberg set for one fixed route (e.g., Cape Town → Bharati, summer) — i.e., an *ice-aware route + risk + explanation* prototype using current / historical ice, with Stage 2 basic (physics + persistence). Self-contained, verifiable and honest.
- **Add second (stronger):** Stage-1 ConvLSTM SIC forecast (trained on public SIC CDR) replacing “known SIC,” with skill-vs-persistence.
- **Add third (advanced):** Stage-2 ensemble iceberg with uncertainty cones; proper POLARIS risk; D* Lite replanning; a multi-objective Pareto UI.
- **Future / optional:** SAR-based SIC / iceberg nowcast; physics-constrained SIC; operational vessel curves; on-ship offline deployment.

Constraints respected: all training / feeds use free open data; every component is decoupled so it can run offline; a single modest GPU (or even CPU training on a regional crop) suffices; no autonomous control; uncertainty and explanation are present throughout.

15 Mathematical Formulation

Notation: cells are indexed by space $\mathbf{s}$ and time t ; distances on a polar grid are great-circle / geodesic.

15.1 Sea-ice concentration and its forecast

Let $C(\mathbf{s}, t) \in [0, 1]$ be the sea-ice concentration. A forecast to lead ℓ is $\hat{C}(\mathbf{s}, t + \ell)$, with an ensemble $\{\hat{C}_k(\mathbf{s}, t + \ell)\}_{k=1}^K$ giving a probabilistic field (mean + spread → confidence). A SIC threshold / the ice-edge contour defines the navigability bands used by routing.

15.2 Iceberg kinematics (drift)

Let $\mathbf{x}_b(t)$ be the iceberg position and $\mathbf{v}_b$ its velocity. A first-order, advection-dominated model is

$$\mathbf{v}_b(t) \approx \mathbf{v}_c(\mathbf{x}_b, t) + \gamma \mathbf{v}_a(\mathbf{x}_b, t) + (\text{Coriolis/geometry correction}), \quad \frac{d\mathbf{x}_b}{dt} = \mathbf{v}_b, \quad (2)$$

where $\mathbf{v}_c$ is the ocean current at the iceberg's keel / draft, $\mathbf{v}_a$ the 10-m wind, and γ an empirical wind-drift factor (≈ 0.01 – 0.035 , depending on size / draft / aerial drag). A full force balance instead integrates $m \mathbf{a} = \sum \mathbf{F}$ (wind + water form / skin drag + Coriolis + surface slope + sea-ice drag). Because $\mathbf{v}_c$, $\mathbf{v}_a$ and γ are uncertain, we run an **ensemble**: sample current / wind/ γ to obtain a set of trajectories $\{\hat{\mathbf{x}}_b(t)\}$, and compute, at each cell and time, the encounter probability

$$p_{\text{berg}}(\mathbf{s}, t) = \frac{\# \{\text{ensemble members within a safety radius of } \mathbf{s} \text{ at } t\}}{K}. \quad (3)$$

15.3 Vessel movement / transit model

Along a candidate path, the attainable speed $u(\mathbf{s}, t)$ is a function of installed power, the ice regime (SIC / thickness / stage) and ice class:

$$u(\mathbf{s}, t) = \mathcal{F}_{\text{vessel}}(P, \text{ice-regime}(\mathbf{s}, t), \text{ice-class}). \quad (4)$$

The transit time per cell is $\Delta t(\mathbf{s}, t) = \Delta d / u(\mathbf{s}, t)$, and fuel is the integral of the engine-power curve,

$$\text{Fuel} = \int \dot{f}(u(\tau), \text{ice}(\tau)) d\tau, \quad (5)$$

where $\dot{f}$ is the fuel rate.

15.4 Route, cost and constraints

A route $R = \{\mathbf{s}_1, \dots, \mathbf{s}_N\}$ is an ordered set of waypoints traversed over time; because the ice moves, cost is integrated along the times the vessel actually occupies each cell (time-dependent).

A scientifically reasonable *balanced* cost is

$$\mathcal{C}(R) = \lambda_{\text{time}} T(R) + \lambda_{\text{fuel}} \text{Fuel}(R) + \lambda_{\text{risk}} \text{Risk}(R), \quad (6)$$

where

$$\text{Risk}(R) = \sum_{\text{along path}} [w_{\text{ice}} \text{ice_risk}(\mathbf{s}, t) + w_{\text{berg}} p_{\text{berg}}(\mathbf{s}, t) + w_{\text{wx}} \text{weather_risk}(\mathbf{s}, t)], \quad (7)$$

subject to hard (no-go) constraints: POLARIS RIO below the permitted limit; avoid land / shallow water; achievable vessel speed; an arrival-time window; and (for a conservative route) an iceberg-encounter probability below a chosen safety threshold.

- $T(R) = \sum \Delta t(\mathbf{s}, t)$ — travel time; $\text{Fuel}(R) = \int \dot{f}$ — fuel.
- ice_risk via a POLARIS-like RIO (ice class $\times$ ice regime / stage $\rightarrow$ risk category).
- p_{berg} from the Stage-2 ensemble; weather_risk penalises extremes beyond ship limits.
- The weights $\boldsymbol{\lambda} = (\lambda_{\text{time}}, \lambda_{\text{fuel}}, \lambda_{\text{risk}})$ encode *route personality*: e.g., large $\lambda_{\text{risk}} \rightarrow$ “safe” route; large $\lambda_{\text{fuel}} \rightarrow$ “economy” route; sweeping $\boldsymbol{\lambda}$ traces a Pareto frontier.

15.5 Multi-objective view and uncertainty in routing

A single weighted sum hides trade-offs and can be sensitive to weight choice, so the scientifically defensible framing presents the **Pareto-optimal set** over objectives (time, fuel, risk) and lets the operator choose. Because the Stage-1 SIC and Stage-2 iceberg forecasts are probabilistic, a robust / risk-aware route either (a) uses mean fields with safety margins, or (b) optimises a risk measure (e.g., chance-constrained: keep $\Pr[\text{besetment or encounter}] \leq \alpha$ along the route). For the MVP, approach (a) — mean fields + margins + displayed uncertainty bands — is appropriate; (b) is an advanced feature. *Equation (6) is illustrative, not the only valid objective; it must be derived and justified for the target vessel and route.*

16 Route-Planning Methodology (Recommended Concrete Pipeline)

1. **Define domain and grid:** the corridor between origin and destination for the season; ~5–10 km cells along the route with a lateral band. Static layers: bathymetry, land, grounding line, protected / zones.
2. **Vessel profile:** choose a reference vessel (e.g., a chartered ice-strengthened vessel or a PC-class research ship) → POLARIS ice class; set permitted ice regimes; set speed–power curves (placeholder if proprietary).
3. **Build time-varying cost layers** over the planned window: a no-go mask (POLARIS RIO, land, shallow, safety buffer, extreme-weather thresholds); an ice-transit / fuel cost as a function of the SIC forecast over time; an iceberg dynamic cost from the Stage-2 encounter-probability field (grown by the vessel turning radius); and a weather cost (wave / wind limits).
4. **Search:** least-cost time-dependent A*/Dijkstra within the allowed cells to produce the recommended route for a chosen λ ; store path, ETA, fuel and cumulative risk.
5. **Alternates:** run λ sweeps / a multi-objective pass (NSGA-III or repeated weighted A*) → a small set of Pareto routes with a trade-off plot (safe vs. fast vs. cheap).
6. **Replanning trigger:** when the Stage-1 / 2 forecasts refresh (e.g., a new daily SIC) or on operator request, recompute incrementally (D* Lite) and flag route changes.
7. **Explain and verify:** annotate why each segment was chosen; show which factors dominate; “hindsail” the chosen route against archived ice to demonstrate skill; log everything.

17 Evaluation Metrics

No accuracy percentages are invented here; metrics follow the SIPN-South / forecast / routing literature, and reported figures in Section 7 speak for the cited papers. For forecasts, the honest standard is *skill relative to persistence / climatology*, not raw RMSE alone.

Sea-ice forecasting (SIC field).

- Pixel MAE / RMSE between forecast and analysis.
- **Anomaly correlation coefficient (ACC)** — the SIPN-standard spatial skill measure.
- **Integrated ice-edge error (IIEE)** — area mismatch of the chosen (e.g., 15%) concentration contour; the most navigation-relevant error (edge position).
- **Binary accuracy / Brier score** for probabilistic (IceNet-style) output.
- **Skill score vs. persistence** (and vs. climatology) — required to claim any skill.
- Qualitative: ice-edge band position vs. observed.

Iceberg prediction.

- **Average / Final Displacement Error (ADE / FDE)** — geodesic distance between predicted and observed position, averaged over the horizon and at the final time (used by IDRIFT-NET and the trajectory literature).
- Position error at fixed leads (6 / 12 / 24 / 48 h, etc.).
- **Uncertainty calibration:** fraction of observed positions inside the ensemble / uncertainty cone (should $\approx$ stated confidence).
- Encounter-level: does the predicted zone contain the observed track within the safety radius?

Route planning.

- Route length / time (geodesic vs. planned) and ETA accuracy.

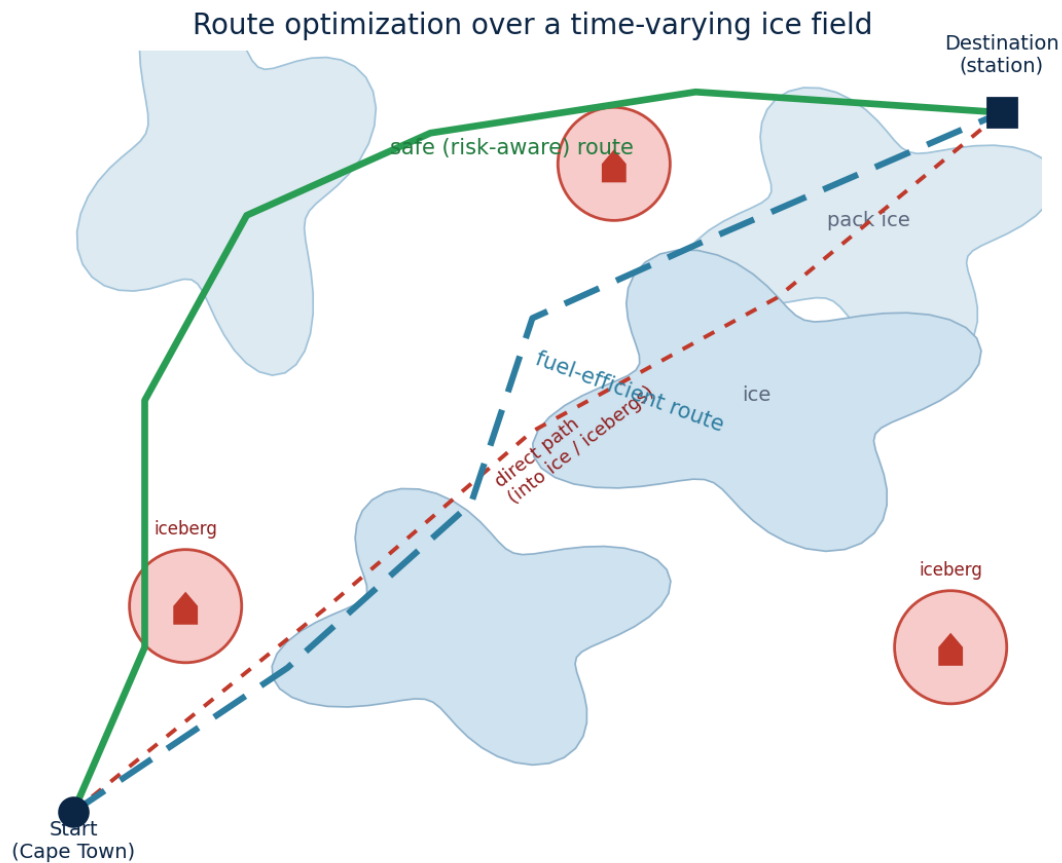


Figure 5: Multi-objective route optimisation over a time-varying ice field (schematic). A naive direct path crosses pack ice and iceberg danger zones; a *risk-aware* route keeps a wide, safe berth (green), while a *fuel-efficient* route skirts the ice (blue) at the cost of higher risk. A decision-support system presents these alternatives and their trade-offs rather than a single “optimal” line. (Conceptual illustration.)

- **Fuel estimate** (needs a calibrated transit model; report as an estimate with caveats; validate if real logs are available).
- **Risk exposure** under the chosen risk model (cumulative RIO / ice / encounter-weighted exposure).
- **Safety-constraint satisfaction** (never violates no-go / POLARIS / encounter threshold).
- **Robustness / uncertainty:** how much the recommended route changes when the SIC / iceberg forecast changes (sensitivity); probability the route remains feasible (besetment-free) under the ensemble.
- Compare against a naive “great-circle in open water, then straight to port” baseline to show added value.

18 Risks and Limitations (Proposed System and Project)

- **Data honesty:** real-time high-resolution currents and winds in / under Antarctic ice, and precise iceberg tracks, are limited; forecasts carry large, growing error. Overstating capability would be operationally dangerous.
- **Validation scarcity:** there are few ground-truth iceberg tracks and vessel logs to verify against, and SIC “truth” itself has error near the edge. The prototype must report this.
- **Physics / scale of decision:** an ice-route decision is the crew’s / authority’s responsibility; a DSS cannot certify safety. Human-in-the-loop must be retained and stated.
- **Component-coupling risk:** forecast errors compound into route errors; uncertainty bands should be shown so the operator sees degraded confidence and the UI does not overclaim.
- **Compute / bandwidth:** onboard constraints dictate an offline / CPU design with periodic sync, not a cloud-only service.
- **Regime shift:** Antarctic ice has shown extreme recent lows (e.g., the 2023 austral-winter record low); models trained on the past may mis-handle new regimes — a real, citable concern (SIPN-South notes the climatology recently broke down). Report persistence / skill in the most recent years.
- **Vessel data proprietary:** speed–power / ice-resistance curves for chartered ships are not public; use clearly labelled placeholders and make the system vessel-parameter-driven.
- **Project-scope risk:** attempting live SIC forecast + SAR iceberg detection + full routing + UI is too much for an SIH build. Scope to an honest MVP (Section 19).

19 SIH MVP / Advanced / Future Scope

19.1 MVP (must work; scientifically defensible; achievable in the hackathon / 3-month window)

- A data pipeline for a chosen route and season from open SIC CDR + ERA5 / ORAS5 + IBCSO.
- An **ice-aware multi-objective route optimizer** (grid, A*/Dijkstra + λ sweep $\rightarrow$ safe and economy routes) using *current* / *historical* SIC, with POLARIS-like no-go constraints and a declared reference vessel.
- A **basic iceberg module:** seed from a public iceberg catalogue; physics drift + ensemble cone; encounter alerts.
- **Risk maps + route-comparison UI** (maps + ETA / fuel / risk + per-segment explanation).
- **Verification:** “hindsail” the recommended route against archived SIC; report ACC / skill / edge-error for any forecast component; honestly report uncertainty.

19.2 Advanced (if time / effort allows)

- A ConvLSTM SIC forecast (replacing “known SIC”) with persistence-baseline skill.
- Probabilistic (ensemble) SIC output; D* Lite incremental replanning.
- A multi-objective Pareto layer (NSGA-III) with an interactive trade-off view.
- A Sentinel-1-based iceberg nowcast (heavy) as an optional demonstration.

19.3 Future research (beyond SIH; needs data / operational collaboration with NCPOR)

Operational near-real-time feeds; vessel-specific resistance / fuel curves; route verification on real NCPOR voyages; on-ship offline deployment; physics-constrained SIC; coupling with an actual dynamical forecast for boundary data; and formal chance-constrained (robust) routing.

20 Recent Research Documents Worth Reading

(Verifiable; 2023–2026 prioritised, foundational marked. Starred = start here.) A short verification note: where an item is cited from a journal record that is still being finalised (e.g., some 2026 *Ocean Eng.* / *ESSD* volumes), open the publisher page before including it in the final project references and attach the DOI you actually confirm.

Sea-ice forecasting / prediction

- * Dong, X., Yang, Q., Nie, Y., Zampieri, L., Wang, J., Liu, J., Chen, D. (2024). “Antarctic sea ice prediction with a convolutional long short-term memory network.” *Ocean Modelling* 190:102386. DOI [10.1016/j.ocemod.2024.102386](https://doi.org/10.1016/j.ocemod.2024.102386). [A] — SIC deep-learning core.
- * Massonnet, F., et al. (2023). “SIPN South: six years of coordinated seasonal Antarctic sea ice predictions.” *Front. Mar. Sci.* 10:1148899. DOI [10.3389/fmars.2023.1148899](https://doi.org/10.3389/fmars.2023.1148899). [A] — community context; ML > dynamical.
- * Dong, X., Nie, Y., Wang, J., et al. (2024). “Deep Learning Shows Promise for Seasonal Prediction of Antarctic Sea Ice in a Rapid Decline Scenario.” *Adv. Atmos. Sci.* 41:1569–1573. DOI [10.1007/s00376-024-3380-y](https://doi.org/10.1007/s00376-024-3380-y). [A]
- Andersson, T.R., et al. (2021). “Seasonal Arctic sea ice forecasting with probabilistic deep learning.” *Nat. Commun.* 12:5124. DOI [10.1038/s41467-021-25257-4](https://doi.org/10.1038/s41467-021-25257-4). [Ar] (IceNet design).
- Wulf, T., et al. (2024). “Pan-Arctic sea ice concentration from SAR and passive microwave.” *The Cryosphere* 18:5277–5300. DOI [10.5194/tc-18-5277-2024](https://doi.org/10.5194/tc-18-5277-2024). [Ar] (supplied Ref. 5).
- Liu, Q., Wang, Y., Zhang, R., Yan, H. (2024). “Incorporating physical constraints in a deep learning framework for short-term daily prediction of sea ice concentration.” *Applied Ocean Research* 148:104007. DOI [10.1016/j.apor.2024.104007](https://doi.org/10.1016/j.apor.2024.104007). [Ar]

Iceberg detection / tracking / drift

- * IDRIFTNET (2025). [arXiv:2507.00036](https://arxiv.org/abs/2507.00036) — hybrid physics + DL iceberg-drift forecast (A23A / B22A). [A] — iceberg-component template.
- * Jiao, K., Fraser, A.D., Lohse, J., Wongpan, P., Adams, C., Bradley, A.C. (2026). “Grounded icebergs around Antarctica: a high-resolution dataset derived from deep learning and Sentinel-1 SAR.” *Earth Syst. Sci. Data* 18:6017–.... DOI [10.5194/essd-18-6017-2026](https://doi.org/10.5194/essd-18-6017-2026). [A]
- * Liu, J., et al. (2026). “A six-year circum-Antarctic icebergs dataset (2018–2023).” *ESSD* 18:147–.... DOI [10.5194/essd-18-147-2026](https://doi.org/10.5194/essd-18-147-2026). [A] — validation / track seeds.
- Redmond, E., et al. (2023). “Mapping the extent of giant Antarctic icebergs with deep learning.” *The Cryosphere* 17:4675–4688. DOI [10.5194/tc-17-4675-2023](https://doi.org/10.5194/tc-17-4675-2023). [A]
- Wesche, C., Dierking, W. (2016). “Estimating iceberg paths using a wind-driven drift model.” *Cold Regions Sci. Technol.* DOI [10.1016/j.coldregions.2016.01.008](https://doi.org/10.1016/j.coldregions.2016.01.008). [A]

- Yulmetov, R., Ralph, F. (2019 / 2021). “Iceberg Drift Forecasting Using Machine Learning.” POAC / C-CORE. [Ar] (hybrid, operational).
- Andersson, L.E., Scibilia, F., Imsland, L. (2019). “An iceberg forecast approach based on a statistical ocean current model.” *Cold Reg. Sci. Technol.* 158:128–... [Ar]
- Sentinel-1 wave-mode iceberg studies (*Remote Sensing of Environment*; 2025 / 2026, Wang et al., incl. IByolo / YOLO). [A]

Ice-aware / polar routing

- * Smith, J.D., et al. (2022–24). “Autonomous Passage Planning for a Polar Vessel” / Polar-Route (BAS). Repository: github.com/antarctica/PolarRoute. [A] — closest existing Antarctic architecture.
- * RouteView 2.0 (2025). “A real-time operational planning system for vessels on the Arctic Northeast Passage.” *Environmental Modelling & Software*. [Ar] (deployed dynamic planner).
- Jeong, S.-Y., et al. (2021). “Ship route planning in Arctic Ocean based on POLARIS.” *Ocean Engineering* 234. [Ar] (risk→route standard).
- Multi-objective ice routing: 3-D ant-colony (*Ocean Eng.* 2022); NSGA-III polar multi-objective (Wang et al., *Ocean Eng.* 2026); improved D* Lite (2026, Arctic). [Ar]
- Ma, K., Wang, L., Yu, M. (2026). “Dynamic Ship Route Planning Based on Fusion of Improved A-Star and Rolling Window Methods.” IEEE CCDC. DOI [10.1109/CCDC69976.2026.11560307](https://doi.org/10.1109/CCDC69976.2026.11560307). [G] (supplied Ref. 3).

Forecast verification / benchmarks

- * Massonnet 2023 (SIPN South, above) — metrics and data. [A]

Operational / user needs / standards

- “Sea-ice information and forecast needs for industry maritime stakeholders.” *Int. J. Digital Earth* (2020). DOI [10.1080/1088937X.2020.1766592](https://doi.org/10.1080/1088937X.2020.1766592). [both] — Antarctic stakeholder needs.
- IMO Polar Code; POLARIS (IACS / ABS) risk indexing. [both]
- “Icebreaking polar class research vessels: new Antarctic fleet capabilities.” *Polar Record* (2021, Cambridge). [A]

Data-source references

- Dorschel, B., et al. (2022). “The International Bathymetric Chart of the Southern Ocean Version 2.” *Scientific Data* 9:275. DOI [10.1038/s41597-022-01366-7](https://doi.org/10.1038/s41597-022-01366-7). (IBCSO v2).
- Zuo, H., et al. (2019). “The ECMWF operational ensemble reanalysis–analysis system for ocean and sea ice (ORAS5).” *Ocean Science* 15:779–808. DOI [10.5194/os-15-779-2019](https://doi.org/10.5194/os-15-779-2019).
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- OSI-SAF SIC CDR (OSI-450 / 430); NSIDC SIC products (NSIDC-0051, NSIDC-0081, NSIDC-0803); NOAA CoastWatch AMSR2.

21 References (Consolidated)

All works cited appear in Section 20. Key primary references confirmed accessible during this research:

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22 Final Recommendation — the Honest, Specific Answer

What exactly would you build for SIH 26059: an “Antarctic Ice-Aware Navigation Decision-Support System for NCPOR Logistics Voyages” that couples three modular, separately testable stages on real open data and one real NCPOR-style route (e.g., **Cape Town** → **Bharati** / **Bharati–Maitri**, austral summer):

- **SIC forecast stage** — a **ConvLSTM** daily sea-ice-concentration forecaster (trained on the OSI-SAF / NSIDC SIC CDR), targeted at the route corridor out to ~7–15 days, always reported as *skill vs. persistence / climatology* (not raw RMSE), output as an ensemble / with uncertainty.
- **Iceberg stage** — a **physics force-balance drift model + ensemble spread**, seeded from public Antarctic iceberg catalogues (Jiao 2026 / Liu 2026 / named-giant tracks), producing probabilistic encounter zones with calibrated cones — not point lines.
- **Routing stage** — a **time-varying, vessel- and ice-class-aware multi-objective route**

optimizer (A*/Dijkstra within a POLARIS-style no-go mask; λ -sweep or NSGA-III Pareto of safe / economy / fast), with **D* Lite replanning** when the SIC / iceberg forecast updates.

- **Decision-support shell** — risk and navigability maps, a recommended route and alternates, ETA / fuel / risk, per-segment explanations, uncertainty, alerts, and “hindsail” verification against archived ice. Human-in-the-loop and offline-capable. **No autonomous control.**

Why: every component is grounded in verifiable research (Sections 8, 20). The SIC deep-learning approach is *justified* — not decorative — because Antarctic ML SIC forecasting now demonstrably beats dynamical models (SIPN South; Dong 2024). Icebergs use physics + ensembles (Wesche 2016; IDRIFTNET) because pure deep learning fails under data scarcity. Routing uses standards-based graph optimization (POLARIS; Jeong 2021; A*/D* Lite; PolarRoute as an architecture reference) because it is safe, explainable and computable — not a reinforcement-learning black box.

What makes it different from existing systems: none of the research systems do **all three in one Antarctic, route-specific, decision-support package** — SIC forecasting lives in climate journals (basin / seasonal), iceberg trajectory in cryosphere / oceanography, and ice routing in naval engineering and mostly the Arctic. The integration + Antarctic specificity + vessel / ice-class awareness + explicit uncertainty + verification is the defensible novelty. **Say that precisely; do not claim “novel AI.”**

What you honestly claim as innovation (suggested wording): *“No accessible, existing system couples a short-lead, route-corridor, probabilistic Antarctic sea-ice forecast with iceberg-drift prediction and vessel / ice-class-aware multi-objective routing under uncertainty for Indian research-vessel logistics. We deliver that integrated, explainable decision-support pipeline, evaluated on real NCPOR-style Cape Town–Bharati / Maitri voyages, with skill and uncertainty reported against standard baselines.”*

Three things that would look strong but are scientifically weak — avoid:

1. “We use AI for autonomous ice navigation” (not requested; unsafe).
2. A demo that draws a route from an unvalidated black-box model, with no uncertainty and no ice-class / POLARIS constraints.
3. Claiming to predict icebergs “accurately” to long horizons (even the best research systems stress how quickly this degrades; show uncertainty instead).

Closing. The scientifically credible, defensible SIH deliverable is an *integrated, explainable, uncertainty-aware Antarctic route decision-support prototype* built on open data, hybrid physics + targeted deep learning, and standards-based route optimization — scoped to an honest MVP, verified by “hindsailing,” and positioned as decision support for the humans who navigate NCPOR voyages.